

Proofs and Problem Solving

Week 5: Complex Numbers and Polynomials

Notes based on Martin Liebeck's

A Concise Introduction to Pure Mathematics

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9. Complex Numbers

9.1 Complex Numbers

Complex numbers are an extension of the real numbers, defined by introducing a new special number called i whose square is defined to be -1 .

Definition 9.1 — Complex Numbers. Define i to be a number such that $i^2 = -1$.

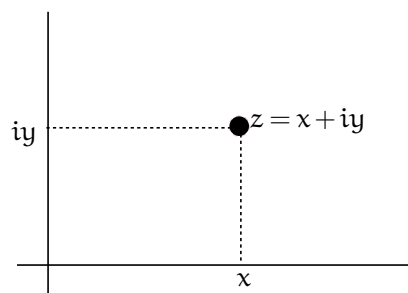
A complex number is any number of the form

$$z = x + iy \text{ where } x, y \in \mathbb{R}.$$

The set of all complex numbers is denoted $\mathbb{C}$.

We define $\operatorname{Re}(z) = x$ and $\operatorname{Im}(z) = y$ to be the *real* and *imaginary parts* of z .

Just like vectors, we represent complex numbers on an xy -plane, where the x -coordinate is the real part and the y coordinate is the imaginary part. Using the plane to represent complex numbers in this way is sometimes called an *Argand diagram*.



Multiplication, addition and subtraction of complex numbers maintain how addition works for real numbers, and treating i as an algebraic letter. In multiplication we proceed

using an obvious approach: consider any two complex numbers $z = a + ib$ and $w = c + id$.

$$\begin{aligned} zw &= (a + ib)(c + id) \\ &= ac + aid + ibc + i^2 bd \end{aligned}$$

This is obtained in the usual way by multiplying all the terms in one bracket by all the terms in the other bracket. Use the defining property of i , which is $i^2 = -1$ so that

$$\begin{aligned} zw &= ac + aid + ibc - bd \\ &= ac - bd + i(ad + bc) \end{aligned}$$

where we have re-grouped terms with the i symbol and terms without the i symbol separately. These are the real and imaginary parts of the product zw respectively.

For example:

$$(5 + 6i)(3 + 4i) = (5 + 3) + (6 + 4)i = 8 + 10i$$

and

$$(1 + 3i)(2 + i) = 1 \cdot 2 + 1 \cdot i + 3i \cdot 2 + 3i \cdot i = 2 + i + 6i - 3 = -1 + 7i.$$

For an integer n , we write $z^n = z \cdot z \cdots z$ as we did with real numbers.

Complex addition can be interpreted as vector addition in the complex plane. Consider any two complex numbers $z = a + ib$ and $w = c + id$.

$$z + w = (a + ib) + (c + id) = (a + c) + i(b + d).$$

This is identical to adding vectors

$$\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a + c \\ b + d \end{pmatrix}$$

What effect does multiplication have?

Example 9.1 We know $z = u + iv \in \mathbb{C}$ is represented in the xy -plane by the point (u, v) . For example, $z = 1 + 5i$ is represented by the point $(1, 5)$ in the plane. As an exercise, plot each of the following complex numbers in the plane.

- | | | | |
|-------------|---------------|---------------|--------------|
| 1. 1 | 2. i | 3. -1 | 4. $-i$ |
| 5. $4 + 3i$ | 6. $-3 + 4i$ | 7. $-3 - 4i$ | 8. $4 - 3i$ |
| 9. $a + ib$ | 10. $-b + ia$ | 11. $-a - ib$ | 12. $b - ia$ |

where you may choose any non-zero a and b you think are suitable.

Multiply each of them by i and plot the result on the same diagram. What do you notice about how this affects the plot of the original complex number? (See Example 9.2)

We will examine the geometry of complex multiplication fully in a short while.

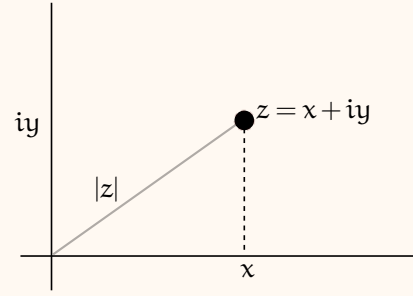
Given $z \in \mathbb{C}$ not equal to zero, we let $z^{-1} = \frac{1}{z}$ denote the complex number such that $z \cdot z^{-1} = 1$. Such a number (i) exists and (ii) is unique (a theorem we omit to save space). In particular, we can divide complex numbers by each other, but before we explain how to do this, there are a couple of important quantities related to a complex number that will make this task easier.

The first is the *modulus*:

Definition 9.2 — Modulus. Given a complex number $z = x + iy$, the *modulus* of z is

$$|z| = |x + iy| = \sqrt{x^2 + y^2}.$$

Geometrically, the modulus of z is its distance from the origin, which is the length of the hypotenuse of a triangle of base x and height y , computed using the Pythagorean theorem.



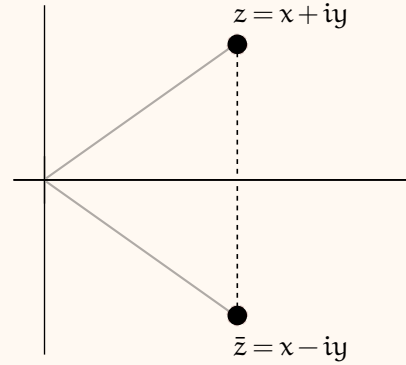
This extends the definition of the absolute value to complex numbers, and in fact, if x is real, the modulus of x is equal to the absolute value. This definition should remind you of the vector norm: for a vector (x, y) , its norm was $\|(x, y)\| = \sqrt{x^2 + y^2}$ as well.

A second important quantity is the conjugate.

Definition 9.3 — Complex Conjugate. Given a complex number $z = x + iy$, its *complex conjugate* is defined to be

$$\bar{z} = x - iy.$$

Geometrically, this is the complex number obtained by reflecting z across the 'x-axis'.



Note: many books and resources use z^* as the complex conjugate, so please get used to using this notation. But choose only one notation in your own writing and stick to it.

One useful thing about a conjugate is

$$z\bar{z} = (a + bi)(a - bi) = a^2 + b^2 = |z|^2.$$

We can use these concepts to show how to divide two complex numbers. E.g. let's look at $\frac{3+i}{1+i}$: we can make the denominator real by multiplying and dividing by the conjugate of $1 + i$ (which is $\overline{1 + i} = 1 - i$):

$$\frac{3+i}{1+i} = \frac{3+i}{1+i} \frac{1-i}{1-i} = \frac{(3+i)(1-i)}{(1+i)(1-i)} = \frac{4-2i}{2} = \frac{4}{2} - i.$$

In general, for $w, z \in \mathbb{C}$ with $z \neq 0$, we have

$$\frac{w}{z} = \frac{w\bar{z}}{z\bar{z}} = \frac{w\bar{z}}{z\bar{z}} = \frac{w\bar{z}}{|z|^2}, \quad \frac{1}{z} = \frac{\bar{z}}{|z|^2} \quad (9.1.1)$$

Here are some other properties that we will use throughout the chapter.

Lemma 9.1 For all $u, v \in \mathbb{C}$:

- $\overline{u+v} = \bar{u} + \bar{v}$.
- $\overline{uv} = \bar{u} \cdot \bar{v}$.
- $|uv| = |u| \cdot |v|$.

We leave the proof as an exercise.

Note (a). $a^2 + b^2$ is “the difference of two squares”

$$a^2 + b^2 = a^2 - (ib)^2 = (a + ib)(a - ib).$$

We can factor expressions over the complex numbers which have no factors over the reals, e.g. $2 = 1^2 + 1^2 = (1 + i)(1 - i)$.

Note (b). The process for finding z^{-1} is very similar to removing surds from the denominator of a rational expression. E.g. compare multiplying $\frac{1}{1+i}$ by the conjugate,

$$\frac{1}{1+i} = \frac{1}{1+i} \frac{1-i}{1-i} = \frac{1-i}{2}$$

with

$$\frac{1}{\sqrt{3}+1} = \frac{1}{\sqrt{3}+1} \frac{\sqrt{3}-1}{\sqrt{3}-1} = \frac{\sqrt{3}-1}{2}.$$

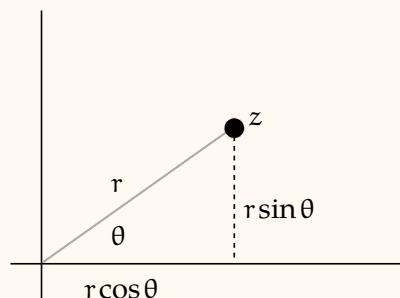
9.2 Cartesian and Polar form

Another useful representation of a complex number $z = x + iy$ is *polar form*. In polar form we represent a complex number by specifying an angle θ from the x axis, and a distance from the origin which, recall, is $|z|$.

Definition 9.4 Let $z \neq 0$ be a complex number. Let $r = |z|$ and define the *argument* of z to be the angle $\theta \in [0, 2\pi)$ between the line from 0 to z and the positive x -axis. We can then write

$$z = r(\cos \theta + i \sin \theta).$$

This is the *polar form* of z .



If we look at the triangle in the above figure, the hypotenuse is $r = |z|$, and so the base and height of the triangle are $r \cos \theta$ and $r \sin \theta$ respectively, and so these give the real and imaginary parts (i.e. the x and y coordinates) of z .

While we specify that the argument θ of a complex number z to be in the range $[0, 2\pi)$, there are infinitely many numbers ϕ so that $z = r(\cos \phi + i \sin \phi)$ (where $r = |z|$): we can just take $\phi = \theta + 2n\pi$ for $n \in \mathbb{Z}$. We have made a choice to only take $\theta \in [0, 2\pi)$. This is called the *principal argument*. Other books and resources sometimes choose $\theta \in (-\pi, \pi]$. Mathematicians are free to “cut” the complex plane where we choose, but be clear about which choice you make and stick to it. For PPS only take $\theta \in [0, 2\pi)$, unless otherwise stated clearly.

9.3 Geometry and arithmetic in $\mathbb{C}$

There are at least three ways to treat complex numbers

1. As a single number z ,
2. As a sum of *real* and *imaginary parts* $z = a + ib$,
3. Using geometry, as a point in the *complex plane*.

Algebra is applied to geometry by representing points in the plane using coordinates. In Cartesian coordinates a point has coordinates (a, b) , so $z = a + ib$. In polar coordinates we have *modulus* r (i.e. length) and *argument* θ (i.e. angle anti-clockwise from the x -axis, or real axis). We can also think of complex numbers as *vectors* from the origin.

One of the most powerful techniques in mathematics is to describe something independently in two different forms and equate them.

Complex numbers have many representations and we will exploit this regularly.

Think about the geometric effect of multiplying $z = x + iy$ by i .

$$i(x + iy) = -y + ix$$

What is the geometric transformation that multiplication by i creates? In vector notation this transformation would be

$$\begin{pmatrix} x \\ y \end{pmatrix} \rightarrow \begin{pmatrix} -y \\ x \end{pmatrix}$$

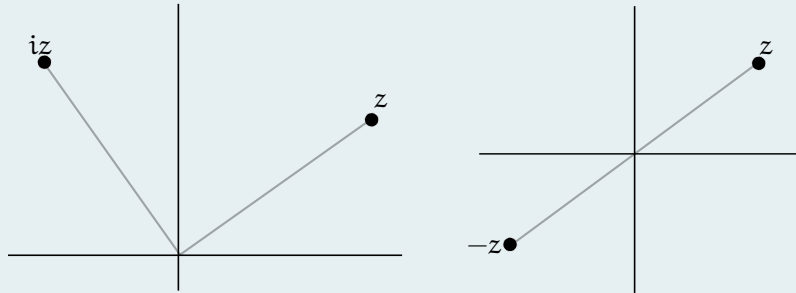
which is an anti-clockwise rotation by $\frac{\pi}{2} = 90^\circ$.

More generally multiplication of complex numbers can be understood by writing the two numbers in polar form. Let $z = r(\cos \theta + i \sin \theta)$ and $w = s(\cos \phi + i \sin \phi)$ then zw

$$\begin{aligned} zw &= r(\cos \theta + i \sin \theta) \times s(\cos \phi + i \sin \phi) \\ &= rs(\cos \theta \cos \phi - \sin \theta \sin \phi) + rsi(\cos \theta \sin \phi + \sin \theta \cos \phi) \\ &= rs(\cos(\theta + \phi) + i \sin(\theta + \phi)). \end{aligned}$$

Notice the overall geometric effect is to just multiply the moduli and add the arguments.

Example 9.2 Since $i = \cos(\pi/2) + i \sin(\pi/2)$, multiplying z by i corresponds to rotating counter clockwise by $\frac{\pi}{2}$, and multiplying by $-1 = \cos(\pi) + i \sin(\pi)$ corresponds to rotating z 180 degrees, i.e. flipping z in the opposite direction.



Given z^n is just z multiplied together n times we can prove De Moivre's Theorem.

Theorem 9.2 — De Moivre's Theorem. If we let $z = r(\cos \theta + i \sin \theta)$, and $n \in \mathbb{N}$ then

$$\begin{aligned} z^n &= r^n(\cos n\theta + i \sin n\theta) \\ z^{-n} &= r^{-n}(\cos(-n\theta) + i \sin(-n\theta)) = \frac{1}{r^n}(\cos(n\theta) - i \sin(n\theta)) \end{aligned}$$

Proof. We prove the first statement by induction. Let $P(n)$ be the statement

$$z^n = r^n(\cos(n\theta) + i \sin(n\theta)).$$

The base case $n = 1$ holds immediately by definition.

Assume $P(n)$ holds for some integer $n \geq 1$.

$$z^{n+1} = zz^n = z \times r^n(\cos(n\theta) + i\sin(n\theta))$$

using the induction hypothesis. And so this

$$\begin{aligned} &= r(\cos \theta + i\sin \theta) \times r^n(\cos(n\theta) + i\sin(n\theta)) \\ &= r^{n+1}(\cos \theta \cos(n\theta) - \sin \theta \sin(n\theta) + i(\cos \theta \sin(n\theta) + \sin \theta \cos(n\theta))) \\ &= r^{n+1}(\cos((n+1)\theta) + i\sin((n+1)\theta)), \end{aligned}$$

which proves $P(n+1)$ holds.

Since $P(1)$ and $P(n) \rightarrow P(n+1)$ it follows that the first equation in the theorem holds by the principle of mathematical induction.

For the second part, we just note that by (9.1.1),

$$\frac{1}{\cos n\theta + i\sin n\theta} = \cos n\theta - i\sin n\theta,$$

and so

$$z^{-n} = (z^n)^{-1} = (r^n(\cos n\theta + i\sin n\theta))^{-1} = r^{-n}(\cos n\theta - i\sin n\theta) = r^{-n}e^{-in\theta}.$$

■

9.4 Exponential form

De Moivre's Theorem allows us to calculate powers z^n for $n \in \mathbb{N}$, but what about more a general exponential such as e^z when $z \in \mathbb{C}$. Euler's formula provides the answer.

Theorem 9.3 — Euler's formula.

$$e^{i\theta} = \cos(\theta) + i\sin(\theta).$$

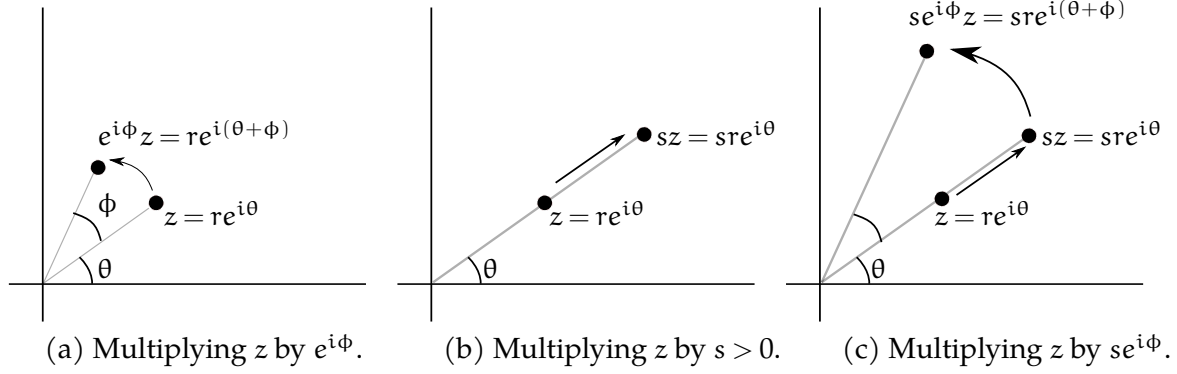
Euler's formula is one of the most remarkable and important formulae in mathematics. E.g. the special case $e^{i\pi} = -1$ links together i (algebra), π (geometry) and e (calculus, since $\frac{d}{dx}e^x = e^x$) in a single equation. Euler's formula justifies the definition of *exponential form* of a complex number z as.

$$z = re^{i\theta} = r(\cos \theta + i\sin \theta). \quad (9.4.1)$$

Exponential notation is a very convenient form for writing complex numbers.

There are many approaches to proving Euler's formula. If you are studying Calculus and Taylor Series, one approach is to show that $e^{i\theta} = \cos \theta + i\sin \theta$ by plugging $i\theta$ into the Taylor series for e^x and then you can split the series into $\cos \theta + i\sin \theta$. We omit this working here, but it is provided as part of the online lecture quiz.

Euler's formula also helps us compute the product of complex numbers, as illustrated in the following figure:



Exponential notation makes taking a power of a complex number much easier if we know its polar form.

Example 9.3 What is $(1+i)^6$?

We could multiply this out by using the Binomial Theorem, but we'll use polar coordinates instead: To write $1+i$ in polar form $re^{i\theta}$, we first find r :

$$r = |1+i| = \sqrt{1^2 + 1^2} = \sqrt{2}.$$

Then

$$e^{i\theta} = \frac{1+i}{r} = \frac{1+i}{\sqrt{2}} = \frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}} = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} = e^{i\frac{\pi}{4}}.$$

Finally,

$$(1+i)^6 = (\sqrt{2})^6 e^{i\frac{6\pi}{4}} = 8 \left(\cos \frac{6\pi}{4} + i \sin \frac{6\pi}{4} \right) = 8 \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right) = -8i.$$

9.5 Roots of Unity

How do we find all solutions to $z^n = 1$?

We now have enough tools in place to answer this. We find them all in a few steps:

- First, using polar coordinates, we can find one root rather easily: $w = e^{\frac{2\pi i}{n}}$, since then

$$w^n = e^{\frac{2\pi i}{n} \cdot n} = e^{2\pi i} = 1.$$

- In particular, this means that any power of w is also a root, since if $j \in \mathbb{N}$,

$$(w^j)^n = (w^n)^j = 1.$$

- Finally, we will show later that for any degree n polynomial there are at most n distinct roots. So if we show that the numbers

$$1, w, w^2, \dots, w^{n-1}$$

are all distinct, then we will have all the roots.

- Suppose for the sake of a contradiction that there are distinct integers j and k with $0 \leq j, k < n$ so that $w^j = w^k$. Then

$$1 = w^{j-k} = e^{(j-k)\frac{2\pi i}{n}} = \cos \left((j-k)\frac{2\pi}{n} \right) + i \sin \left((j-k)\frac{2\pi}{n} \right).$$

The only way this can be 1 is if the cosine is 1 and the sine is zero, so which only happens if their arguments are multiples of 2π , that is, we must have

$$(j-k)\frac{2\pi}{n} = 2\pi\ell \text{ for some integer } \ell$$

which implies $j-k = n\ell$ for some $\ell \in \mathbb{Z}$, but this is impossible since $0 \leq j, k < n$, and hence $-n < j-k < n$, so $j-k$ must be zero, and so j and k cannot be distinct. Thus, the numbers $1, w, \dots, w^{n-1}$ are distinct and form all the roots.

We have thus shown the following.

Theorem 9.4 — Roots of Unity. The solutions to $z^n = 1$ are $1, w, \dots, w^{n-1}$ where $w = e^{\frac{2\pi i}{n}}$. That is, they are $e^{\frac{2\pi ki}{n}}$ for $k = 0, 1, \dots, n-1$.

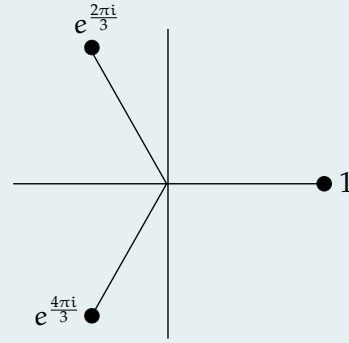
Example 9.4 — The third roots of unity.

The third roots of unity are

$$z_1 = 1$$

$$z_2 = e^{2\pi i/3} = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} = -\frac{1}{2} + i \frac{\sqrt{3}}{2}$$

$$z_3 = e^{4\pi i/3} = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} = -\frac{1}{2} - i \frac{\sqrt{3}}{2}$$



Note: the cube roots of unity are used so frequently that writers often define $\omega := e^{2\pi i/3}$. Then we have $\omega^2 = \bar{\omega}$, and lots of other interesting and useful algebraic relationships.

How do we find all solutions to $z^n = a$ where $a \in \mathbb{C}$?

Again, we can use n th roots of unity. We showcase the method in an example:

Example 9.5 Suppose we set $a = 16i$ and wish to find the fourth roots of a (i.e., solve $z^4 = 16i$). We can find one quick root using polar coordinates: note that $a = 16e^{i\pi/2}$, so we can spot one root as

$$z = 16^{1/4} e^{i\frac{\pi}{2} \cdot \frac{1}{4}} = 2e^{i\frac{\pi}{8}},$$

since then $z^4 = a$.

Recall that there are at most 4 distinct roots for a degree 4 polynomial. Note that since z is a solution, if $1, w, w^2, w^3$ (that is, $1, e^{i2\pi/4} = e^{i\pi/2}, e^{i4\pi/4} = e^{i\pi}$, and $e^{i6\pi/4} = e^{i3\pi/2}$) are the 4th roots of unity, then for $k = 0, 1, 2, 3$,

$$(zw^k)^4 = z^4 w^{4k} = a \cdot 1 = a.$$

Thus, the other solutions are z, zw, zw^2, zw^3 . For $k \in \{0, 1, 2, 3\}$ this gives the following (in exponential form)

$$2e^{i\frac{\pi}{8}}, 2e^{i\frac{5\pi}{8}}, 2e^{i\frac{9\pi}{8}}, 2e^{i\frac{13\pi}{8}}$$

or equivalently

$$2e^{i\frac{\pi}{8}}, 2e^{i\frac{5\pi}{8}}, 2e^{i\frac{-7\pi}{8}}, 2e^{i\frac{-3\pi}{8}}$$

9.6 Exercises

The relevant exercises in Liebeck's book are in Chapter 6.

■ **Exercise 9.1** Show that $\Re(z) = \frac{z+\bar{z}}{2}$ and $\Im(z) = \frac{z-\bar{z}}{2i}$.

■ **Exercise 9.2** Show that

$$\Re(zw) \leq \frac{|z|^2 + |w|^2}{2}.$$

■ **Exercise 9.3** Solve $z^2 = i\bar{z}$.

■ **Exercise 9.4** Solve $|z|^2 - z|z| + z = 0$.

■ **Exercise 9.5** Show that if $|z| = 1$, then $\Re\frac{1}{1-z} = \frac{1}{2}$.

■ **Exercise 9.6** Describe geometrically the points $z \in \mathbb{C}$ so that $|z-1| = |z+i|$.

■ **Exercise 9.7** Find all solutions to $(z+1)^4 = z^4$.

■ **Exercise 9.8** Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$, that is, the set of complex numbers with modulus strictly less than 1. $\mathbb{D}$ is for “disc”. For a complex number $z \in \mathbb{D}$ and a real number $-1 < r < 1$, define¹

$$f_r(z) = \frac{z-r}{1-zr}.$$

Show $f_r(z) \in \mathbb{D}$ for all $z \in \mathbb{D}$ and $-1 < r < 1$.

¹The function f_r is called a *Möbius transformation* and is usually defined for $r \in \mathbb{D}$ as well, not just $-1 < r < 1$. They are the only functions with the property that, if A is a circle or straight line in $\mathbb{D}$, then $f_r(A)$ is also a line or circle (this is not part of the problem, it's just cool).

10. Polynomials

10.1 Introduction

Definition 10.1 For $n \in \mathbb{N}$, an n -degree complex polynomial p is a function of the form

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0 \quad (10.1.1)$$

where $a_n \neq 0$ and $a_i \in \mathbb{C}$ for all i .

n is called the *degree* of the polynomial.

A *root* of p is a complex number $\alpha \in \mathbb{C}$ such that $p(\alpha) = 0$.

If the degree of a polynomial is 2 then the polynomial is called a *quadratic*. Any quadratic $az^2 + bz + c$ has two roots which can be found using the familiar *quadratic formula*:

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Note: One needs to be careful here, since we are allowing $a, b, c \in \mathbb{C}$, so $b^2 - 4ac$ could be complex. In this case, $\sqrt{b^2 - 4ac}$ is interpreted to mean any number z so that $z^2 = b^2 - 4ac$, and then the solutions are $\frac{-b \pm z}{2a}$.

There are also formulae for the roots of cubic or quartic (i.e. degree 3 or 4) polynomials, although these are much less convenient to write down. It is natural to ask whether there are convenient formulae for higher order polynomials, but this is not the case:

Theorem 10.1 — Abel-Ruffini Theorem. There is no formula (like the quadratic formula) for the roots of a polynomial of degree ≥ 5 .

This doesn't mean degree 5 polynomials don't have roots, we will see below that they must have at least one root. This also doesn't mean it is impossible to solve higher order polynomials for their roots, it just means we do not have an expression (i.e. formula) for the roots of the polynomial in terms of the coefficients.

10.2 Factorizing Polynomials

The following theorem is stated without proof in PPS (although we prove it later in the degree):

Theorem 10.2 — Fundamental Theorem of Algebra. Any complex polynomial has at least one root in $\mathbb{C}$.

Using this, we can prove that any polynomial can be factored completely.

Theorem 10.3 — Factorization Theorem. If p is a degree n polynomial, then there are n roots $r_1, \dots, r_n \in \mathbb{C}$ and a number $a \in \mathbb{C}$ so that

$$p(z) = a(z - r_1)(z - r_2) \cdots (z - r_n).$$

In the above theorem some roots may repeat. If a root appears m times in $r_1, \dots, r_n$, then we say it has *multiplicity* m .

Notice that if we multiply out the above polynomial, then we get a polynomial like $p(z) = az^n + \dots$, so if p was as in (10.1.1), then we actually know that $a = a_n$.

The above theorem tells us that an n -degree polynomial always has n roots if we also take into account the multiplicity of the roots (that is, how often it appears in the factorization above). For example, $(z - 1)^2$ is a 2-degree polynomial with only one root, but that root has multiplicity 2.

Proof. We prove this by induction on n , the degree of the polynomial p . Let $P(n)$ be the statement

A degree n polynomial p has n roots $r_1, \dots, r_n \in \mathbb{C}$ and there is a number $a \in \mathbb{C}$ so that

$$p(z) = a(z - r_1)(z - r_2) \cdots (z - r_n).$$

Base Case: Assume p is any degree 1 polynomial. If we write p as $p(z) = az + b$, we can re-write $p(z) = a(z - \frac{-b}{a})$. Then p has precisely one root $r_1 = \frac{-b}{a}$ and p can be written in the form $p(z) = a(z - r_1)$. This proves $P(1)$.

Induction Step: Suppose the theorem holds true for some integer $n \geq 1$, i.e. $P(n)$ is true. Let p be a degree $n + 1$ polynomial. By the Fundamental Theorem of Algebra there is a root w of p , i.e. $p(w) = 0$.

Consider the polynomial $q(z) := p(z + w)$. Then q is also degree $n + 1$ and has a root at 0. Write q out as a polynomial then because it has a root at $z = 0$ the constant term is also zero and we can write

$$q(z) = q_1z + \cdots + q_{n+1}z^{n+1} = z \underbrace{(q_1 + q_2z + \cdots + q_{n+1}z^n)}_{=g(z)}.$$

where g is a polynomial of degree n . By our induction hypothesis $P(n)$, there are $a, z_1, \dots, z_n \in \mathbb{C}$ so that

$$g(z) = a(z - z_1)(z - z_2) \cdots (z - z_n).$$

Note if $q(z) := p(z + w)$ then

$$p(z) = q(z - w) = (z - w)g(z - w)$$

and so if $r_i := z_i + w$ for $i = 1, \dots, n$ and $r_{n+1} = w$ then

$$p(z) = a(z - w - z_1)(z - w - z_2) \cdots (z - w - z_n)(z - w) = a(z - r_1)(z - r_2) \cdots (z - r_{n+1}).$$

This proves p has $n+1$ roots and there is a number $a \in \mathbb{C}$ so that we can write

$$p(z) = a(z-r_1)(z-r_2)\cdots(z-r_{n+1}).$$

Since $P(1)$ is true, and $P(n) \Rightarrow P(n+1)$ it follows that $P(n)$ is true for all $n \in \mathbb{N}$ by the principle of mathematical induction. ■

10.3 Real Polynomials

If we are given a *real polynomial*, by which we mean a polynomial $p(z) = a_0 + \cdots + a_n z^n$ such that each $a_i \in \mathbb{R}$, then we have a bit more information about the roots:

Theorem 10.4 — Real Polynomials have conjugate roots. If $p(x)$ has *real* coefficients and r is a root, so is $\bar{r}$.

Proof. If $p(x) = a_0 + a_1 x + \cdots + a_n x^n$ with a_i real, and $p(r) = 0$, then

$$\begin{aligned} 0 &= \overline{p(r)} \\ &= \overline{a_0 + a_1 r + \cdots + a_n r^n} \\ &= \overline{a_0} + \overline{a_1 r} + \cdots + \overline{a_n r^n} && \text{(applying } \overline{z+w} = \bar{z} + \bar{w} \text{ for all } z, w \in \mathbb{C}) \\ &= \overline{a_0} + \bar{a_1} \bar{r} + \cdots + \bar{a_n} \bar{r}^n && \text{(applying } \overline{zw} = \bar{z}\bar{w} \text{ for all } z, w \in \mathbb{C}) \\ &= a_0 + a_1 \bar{r} + \cdots + a_n \bar{r}^n && \text{(applying that the } a_i \text{ are real, which implies } \bar{a_i} = a_i) \\ &= p(\bar{r}). \end{aligned}$$

Thus, $\bar{r}$ is a root. ■

Note: If any of the coefficients is not a real number then all bets are off! That is, we won't be able to conclude that conjugates of roots are also roots.

Example 10.1 Find the roots of $x^4 + 2x^3 - 7x^2 + 2x - 8$ given that one of them is i .

Recall that the roots come in conjugate pairs, and so $-i$ is also a root, hence $(x-i)(x+i) = x^2 + 1$ is a factor in the above polynomial, so we can do polynomial long division to see how it factors: first, since the leading term of the above polynomial is x^4 , we subtract a multiple of $x^2 + 1$ that will eliminate the x^4 , so we subtract $x^2(x^2 + 1)$ from the polynomial to get

$$2x^3 - 8x^2 + 2x - 8$$

Now the leading term is $2x^3$, so we remove $2x(x^2 + 1)$ from this to get

$$-8x^2 - 8 = -8(x^2 + 1).$$

Finally, we can eliminate this term by subtracting $-8(x^2 + 1)$. Thus, adding together all the multiples of $(x^2 + 1)$ we subtracted, we get that

$$x^4 + 2x^3 - 7x^2 + 2x - 8 = (x^2 + 1)(x^2 + 2x - 8)$$

Now we just need to solve $x^2 + 2x - 8 = 0$. Using the quadratic formula, we find that the other two roots are 2 and -4 . Thus, all the roots are $\pm i, 2$, and -4 .

Notice how in that example, we started off just knowing one root and from that the polynomial collapsed and we could find the other 3, thus, even with partial information about the roots of a polynomial, we can use reasoning like this to solve for them all.

10.4 Root-Coefficient Theorem

Another useful tool is the following theorem which shows how the coefficients of a polynomial relate to the roots.

Theorem 10.5 — Root-Coefficient Theorem. If $p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0$, has roots $r_1, \dots, r_n$ (counting multiplicities), then

$$r_1 + \cdots + r_n = -a_{n-1}$$

$$r_1 \cdots r_n = (-1)^n a_0.$$

In general, if s_j denotes the sum of all products of j -tuples of the roots (e.g. $s_2 = r_1r_2 + r_1r_3 + r_2r_3 + \cdots$), then

$$s_j = (-1)^j a_{n-j}.$$

A “tuple” is a finite ordered list of items, here the roots. The phrase “ j -tuples” means a list of items of length j .

Proof. First, factorize

$$p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0 = a(x - r_1) \cdots (x - r_n).$$

Note that as the coefficient of $x^n = 1$, we know $a = 1$ (since otherwise the right side, when multiplied out, wouldn't equal the left). We can establish the formulas in the theorem now by multiplying out the product on the right. ■

This theorem looks complicated, but in fact it just encodes a pattern.

$$(x - \alpha) = x - \alpha$$

$$(x - \alpha)(x - \beta) = x^2 - (\alpha + \beta)x + \alpha\beta$$

$$(x - \alpha)(x - \beta)(x - \gamma) = x^3 - (\alpha + \beta + \gamma)x^2 + (\alpha\beta + \beta\gamma + \alpha\gamma)x - \alpha\beta\gamma$$

In the language of the theorem, a single 2-tuple from the set $\{\alpha, \beta, \gamma\}$ would be a two of these items. The set of all different 2-tuples is

$$\alpha\beta, \beta\gamma, \alpha\gamma$$

Example 10.2 Suppose $x^3 + ax^2 + bx + c$ has roots α, β , and γ . Find a polynomial with roots $\alpha\beta, \beta\gamma$, and $\gamma\alpha$ in terms of a, b, c (that is, the coefficients of your polynomial should only be described using a, b , and c , not α, β , and γ).

By the Root Coefficient Theorem,

$$\alpha + \beta + \gamma = -a,$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = b$$

and

$$\alpha\beta\gamma = -c.$$

Let $x^3 + Ax^2 + Bx + C$ be a polynomial with roots $\alpha\beta$, $\beta\gamma$ and $\gamma\alpha$. Then we know

$$-A = \alpha\beta + \beta\gamma + \gamma\alpha = b$$

$$B = \alpha\beta^2\gamma + \alpha\beta\gamma^2 + \alpha^2\beta\gamma = -c(\alpha + \beta + \gamma) = ac$$

and

$$-C = \alpha\beta \cdot \beta\gamma \cdot \gamma\alpha = (\alpha\beta\gamma)^2 = c^2$$

Hence, the polynomial is

$$x^3 - bx^2 + acx - c^2.$$

Example 10.3 Are all the roots of $x^3 + 11x^2 + 7$ integers?

Suppose they were. Notice that by the Factorization Theorem, this polynomial has three roots a, b, c (where some of these roots could repeat). By the Root-Coefficient Theorem, $abc = -7$, so $7 = |abc| = |a| \cdot |b| \cdot |c|$. Since 7 is prime and we are assuming the roots are integers, the only way this is possible is if one of these absolute values is 7 and the others are 1, say $|a| = 7$ and $|b| = |c| = 1$. The Root-Coefficient Theorem also says $a + b + c = -11$, so by the triangle inequality,

$$11 = |-11| = |a + b + c| \leq |a| + |b| + |c| = 1 + 1 + 7 = 9,$$

which is impossible. Thus, at least one of the roots is not an integer.

10.5 Exercises

■ **Exercise 10.1** Find the (complex) roots of the following polynomials:

(a) $x^2 - 5x + 7 - i = 0$.

(b) $x^4 - x^2 - 1 = 0$.

(c) $2x^4 - 4x^3 + 3x^2 + 2x - 2$, given that one of the roots is $1 + i$.

■ **Exercise 10.2** Factor the following polynomials into products of real polynomials that are linear and/or quadratic:

- $x^3 - 1$.
- $x^3 + 1$.
- $x^4 - 1$.
- $x^4 + 1$.
- $x^5 + 1$. *Hint:* $\cos \frac{2\pi}{5} = \frac{-1+\sqrt{5}}{4}$ and $\cos \frac{4\pi}{5} = \frac{-1-\sqrt{5}}{4}$.

■ **Exercise 10.3** If $x^3 + 15x^2 + 74x + 120$ has roots of the form $a, a + 1, a + 2$, find a .

■ **Exercise 10.4** Let $a \in \mathbb{R}$. If $x^3 - x + a$ has three integer roots, solve for a .

■ **Exercise 10.5** Show that if $z_1, z_2, z_3 \in \mathbb{C}$ are so that $z_1 + z_2 + z_3 = 0$, and $z_1^2 + z_2^2 + z_3^2 = 0$ then $|z_1| = |z_2| = |z_3|$

■ **Exercise 10.6** Suppose $|z_1| = |z_2| = |z_3| = 1$ and $z_1 + z_2 + z_3 = 0$. Show that $z_1^3 = z_2^3 = z_3^3$. *Hint: First think about when $z_1 = 1$, then use that to prove the general case.*

■ **Exercise 10.7** Suppose $|z_1| = |z_2| = |z_3| = 1$ and $z_1 + z_2 + z_3 = 0$. Show that $z_1^{2^n} + z_2^{2^n} + z_3^{2^n} = 0$ for all $n \in \mathbb{N}$.